

ACKNOWLEDGMENT

The internship opportunity I had with Oneknotone Technologies was a great chance for learning and professional development. Therefore, I consider myself as a very lucky individual as I was provided with an opportunity to be a part of it.

Bearing in mind previous I am using this opportunity to express my deepest gratitude and special thanks to the CEO of Oneknotone Technologies, Mr. Alap Acharya, who in spite of being extraordinarily busy with his duties, took time out to hear, guide and keep me on the correct path and allowing me to carry out my project at their esteemed organization and extending during the training.

I express my deepest thanks to Mr. Pranav Shukla, Tech Lead for taking part in useful decision & giving necessary advices and guidance and arranged all facilities to make this internship easier. I choose this moment to acknowledge his contribution gratefully.

I would also like to thank my internal guide, Prof. Mala H. Mahta, for his support in the completion of my internship successfully.

Sincerely,

Vineet Murlidhar Dairashri

ABSTRACT

The 3 month internship in Oneknotone Technologies starting from 8th February 2023 8th May 2023 included implementing Artificial Intelligence solutions to facilitate the company's primary product. This included finding potential areas in the field which can be improved using Artificial Intelligence.

This included creating a recommendation system that aids the process of selecting tiles by taking user's likes into consideration and suggesting the most appropriate suggestion for a particular user. This ultimately made the experience more interactive and personalized for each user.

The system was built using python as a programming language to implement Machine Learning models along with an unstructured online database to connect the back-end with the Virtual Reality system. The script was hosted on AWS webserver to make it more accessible. The features of the system includes recommending and taking feedback from user to make their experience simpler.

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LIST OF ABBREVIATIONS

AI	Artificial Intelligence
ML	Machine Learning
VR	Virtual Reality
AWS	Amazon Web Services
KNN	K Nearest Neighbor
UI	User Interface
UX	User Experience

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CHAPTER 1 OVERVIEW OF COMPANY

1.1 HISTORY

Oneknotone Technologies LLP is a Limited Liability Partnership firm incorporated on 02 March 2020.

Oneknotone can be categorized as a tech company which specializes in creating software to ease the daily operations of an enterprise, But OneKnotone is more than that, OneKnotone believes in the idea of creating a future where technology provides innovative solutions for businesses to prosper no matter the challenges ahead.

Onknotone started with the idea of making the workplace more efficient with the implementation technologies that favour the ease of doing business and creating productivity throughout their business cycle, with this in mind they create software and experiences that boost the efficiency of the business.

At Oneknotone Technologies, they use the best of Augmented, Virtual and Mixed Reality technology to solve problems and fill the gaps in daily processes. They believe that Immersive Visual technology is the key to changing the way one learns, works, grows and experiences within our environment. With their values rooted in client satisfaction and the ability to dream, our mission is to provide the best services and products in our industry segment.

1.2 DIFFERENT PRODUCT / SCOPE OF WORK

Oneknotone Technologies is a full service Next Gen Tech Startup. Our flagship solution has been a full immersion VR tour for Real Estate and Industries. We create powerful experiences of enterprise quality that add value to processes under any vertical. Our solutions are produced at AAA quality for best visualization.

1.3 ORGANIZATIONAL CHART

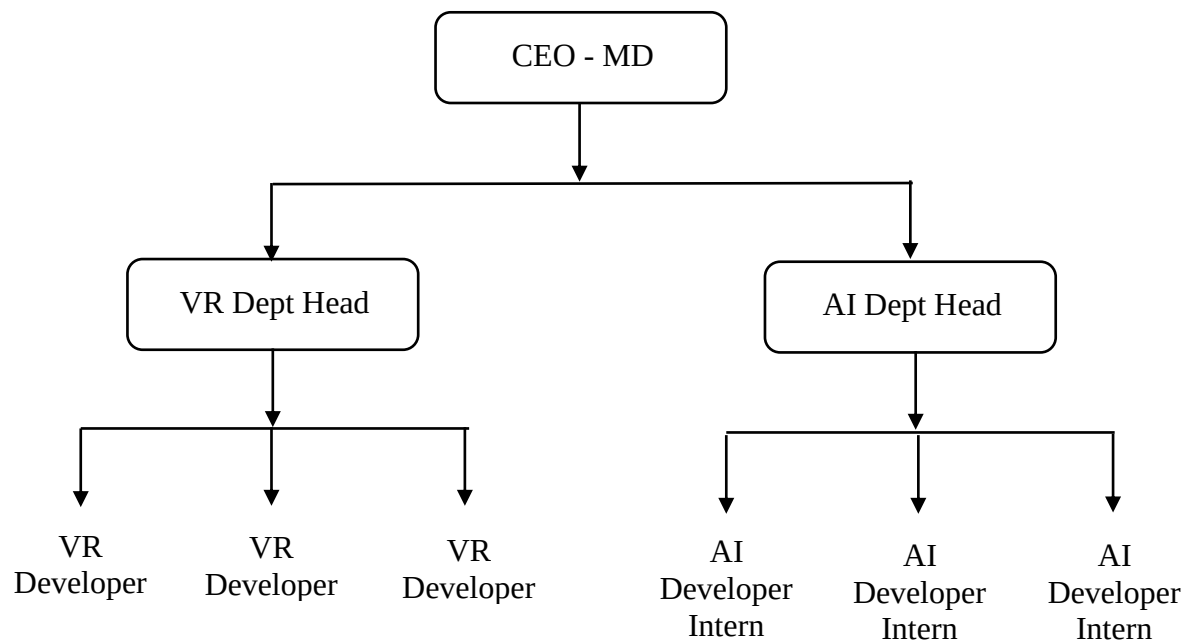


Fig 1.1 Organizational Chart

CHAPTER 2 INTRODUCTION TO INTERNSHIP

2.1 INTERNSHIP SUMMARY

A 3-month internship as a AI Intern, starting from 8th February 2023 till 8th May 2023. In the mode of internship was work from office working closely with VR team to understand their basic product and its working.

The working days were from Monday to Saturday, where the working hours were from 11am to 7pm.

2.2 PURPOSE

The purpose of this internship was to satisfy the requirements towards the completion of the Bachelor of Engineering degree in Information Technology, which requires industrial experience of at least 12 weeks, dedicated and authenticate work in the domain along with getting real world experience of the industry.

2.3 OBJECTIVE

The objective of the internship included creating a recommendation system that aids the process of selecting tiles by taking user's likes into consideration and suggesting the most appropriate suggestion for a particular user. This ultimately made the experience more interactive and personalized for each user.

The system was built using python as a programming language to implement Machine Learning models along with an unstructured online database to connect the back-end with the Virtual Reality system. The script was hosted on AWS webserver to make it more accessible. The features of the system includes recommending and taking feedback from user to make their experience simpler.

2.4 SCOPE

The scope of the internship is to implement AI solution to facilitate the company's products. The scope includes study of the database, designing & coding the new system according to the needs mentioned in the project definition, connecting the backend with the frontend, and then testing the system to see if it able to present the accurate data in acceptable performance rates.

2.5 TECHNOLOGY AND LITERATURE VIEW

The project revolves around providing a AI solution for recommendation of products. It includes proper and easy feedback of the user as well as providing a bridge between the frontend and the backend. The new system is built keeping in my mind user as the primary concern. To provide them easy and accurate recommendation to help them enhance their experience in selection. This includes showing them products that are most suitable for them. We implemented our basic AI program by using KNN model.

The project developed in the internship mainly stands on these technologies, which are Python, Firebase, AWS.

Python :

Python is a popular programming language.

- Python can be described as an Interpreted language, which is, in simple terms, implies that it doesn't require compilation into machine language instruction prior to execution, and it is able to be utilized by the programmer directly to execute the program.
- Python is also considered to be a general-purpose programming language, which means that it is able to be utilized across technology and domains.
- Python comes with Prebuilt Libraries such as Numpy to perform scientific calculations, Scipy for advanced computing, and Scikit Learn for machine learning (Python Machine Learning), making it among the top languages for AI.
- Python is platform-independent and therefore is among the most adaptable and well-known options for various platforms and technologies, with minimal modifications to the basics of coding

Firestore :

Google Firestore is a Google-backed application development software that enables developers to develop iOS, Android and Web apps. Firestore provides tools for tracking analytics, reporting and fixing app crashes, creating marketing and product experiment. Firestore offers a number of services, including:

- Analytics – Google Analytics for Firestore offers free, unlimited reporting on as many as 500 separate events. Analytics presents data about user behavior in iOS and Android apps, enabling better decision-making about improving performance and app marketing.
- Authentication – Firestore Authentication makes it easy for developers to build secure authentication systems and enhances the sign-in and onboarding experience for users. This feature offers a complete identity solution, supporting email and password accounts, phone auth, as well as Google, Facebook, GitHub, Twitter login and more.
- Cloud messaging – Firestore Cloud Messaging (FCM) is a cross-platform messaging tool that lets companies reliably receive and deliver messages on iOS, Android and the web at no cost.
- Realtime database – the Firestore Realtime Database is a cloud-hosted NoSQL database that enables data to be stored and synced between users in real time. The data is synced across all clients in real time and is still available when an app goes offline.
- Crashlytics – Firestore Crashlytics is a real-time crash reporter that helps developers track, prioritize and fix stability issues that reduce the quality of their apps. With crashlytics, developers spend less time organizing and troubleshooting crashes and more time building features for their apps.
- Performance – Firestore Performance Monitoring service gives developers insight into the performance characteristics of their iOS and Android apps to help them determine where and when the performance of their apps can be improved.
- Test lab – Firestore Test Lab is a cloud-based app-testing infrastructure. With one operation, developers can test their iOS or Android apps across a variety of devices and device configurations. They can see the results, including videos, screenshots and logs, in the Firestore console.

AWS :

The Amazon Web Services (AWS) platform provides more than 200 fully featured services from data centers located all over the world, and is the world's most comprehensive cloud platform. Amazon web service is an online platform that provides scalable and cost-effective cloud computing solutions. AWS is a broadly adopted cloud platform that offers several on-demand operations like compute power, database storage, content delivery, etc., to help corporates scale and grow.

2.6 INTERNSHIP PLANNING**2.6.1 Internship Development Approach and Justification**

The project was developed in stages where the first stage was learning about the technologies that are already being used, the current system and the requirements for the new system to be implemented. The second stage was coding and designing of the new system. The last stage was to test the system to see if it was working properly. The progress of work was examined by industry mentor on a timely basis with the help of regular meetings and demos.

2.6.2 Internship Efforts and Time, Cost Estimation

The internship efforts, time and cost estimation was calculated according the monthly basis. It includes daily hours worked, leaves taken and overtime.

2.6.3 Roles and Responsibilities

The role for the intern was of a AI Intern and oversaw multiple responsibilities for the project. I was the AI team leader in charge of planning and managing the whole project and the team.

The responsibility of the intern includes:

1. Studying the current system and finding potential areas where AI can be used to increase the efficiency or the experience.
2. Identifying the path of development and managing the whole project while communicating with other teams.
3. Study the working of basic product and get a sense of its working from a technical point of view.
4. Code the required modules to get the project working as required.
5. Connect the developed project with existing project.

6. Testing the code to see if the project responds as required.
7. Refactor the code to remove unnecessary code and refine the code so that there are no inconsistencies.

2.6.4 Group Dependencies

The project was dependent on the existing project implemented by the VR team that allowed a user to view and change design of tiles and other decorative items in Virtual Reality environment using VR glasses. Our role was to make the selection process easier and more personalized.

2.7 PROJECT PLANNING

Overview of Project Planning:

1. The first 2 weeks was for introductions and getting familiar with the previous system and learning the technologies that were required in the new system.
2. The designs for the new system were planned.
3. Started developing a program to scrap data from website to be used as our dataset.
4. After collection of data it was analyzed for inconsistency and then after modified.
5. Creating a basic KNN model that gives us the desired results .
6. Learning Firebase and its connectivity to python.
7. Connecting the output of our model to the main product using Firebase.
8. Learning AWS to host our script.
9. Hosting the code in a server to make it more accessible.
10. Testing the product for unusual behavior and making appropriate changes.
11. Planning various other AI models and repeating the whole process again while checking their performance for the given dataset.

CHAPTER 3 SYSTEM ANALYSIS

3.1 STUDY OF CURRENT SYSTEM

The current system is made on the technologies like Unreal Engine, Virtual Reality, Mixed Reality, Firebase. The working is very straight forward. Users can surf through the products provided using voice commands or manually entering the product number.

By doing this the users can change almost all aspects of an area and visualize it at the same time using Virtual Reality glasses.

The process of presenting a product design required an expert operator who had knowledge of all designs. If a user needed counseling for which course to choose then that was provided by the operator based on his intelligence. This was not at all an efficient way to show user product designs.

The current system was not user friendly, this was deducted by the feedback received from users. To suggest better products and make the whole process independent we planned on using AI.

3.2 PROBLEM AND WEAKNESS OF CURRENT SYSTEM

Given different and wide-ranging benefits of the current system, there are certain setbacks.

The limitations of the current system are:

1. The product required experienced operator who had knowledge of the system in depth which is not always possible.
2. The current system provided voice commands to change designs but remembering each product code was not the best way.
3. Lack of proper user analytical and user friendly dashboard that displayed the information of the product such as its name, price and design.
4. Users can only view the products and not like or dislike it to get more suitable recommendations.

3.3 REQUIREMENTS OF THE NEW SYSTEM

In the light of the weaknesses of the current system, the requirements of the new system are as follows:

1. Totally automating the experience and removing the need of a professional.
2. Better system to get user inputs for better understanding the user.
3. A proper dashboard where the users can get all the required information of a product.
4. Getting user like and dislike to understand their behavior towards a product.

3.4 PROPOSED SYSTEM

The proposed system is building a fully functional recommendation system using ML models. This project includes collecting and analyzing data to understand it better. It also includes building a system that recommends products based on user likes. During the span of the internship, considerable time was spent on collecting data and learning about the technology that was used to connect the script to the database, coding and designing the new system.

The new system will have the following features:

1. Totally automating the experience and removing the need of a professional.
2. Better system to get user inputs for better understanding the user.
3. A proper dashboard where the users can get all the required information of a product.
4. Getting user like and dislike to understand their behavior towards a product.

3.5 FEATURES OF NEW SYSTEM

The features of the new system are:

1. The system will have almost no external help required because of the simple UI and its automated recommendation.
2. Better recommendation for each user based on their personal preferences.
3. A proper dashboard where the users can check the details of the product easily.
4. Users can like a product to see more similar products.
5. A detailed and well structured description of the products. Includes the price, details of the modules and other necessary information.

CHAPTER 4 SYSTEM DESIGN

4.1 SYSTEM DESIGN AND METHODOLOGY

The system is designed and implemented in the Agile Methodology.



Fig 4.1 Agile Methodology

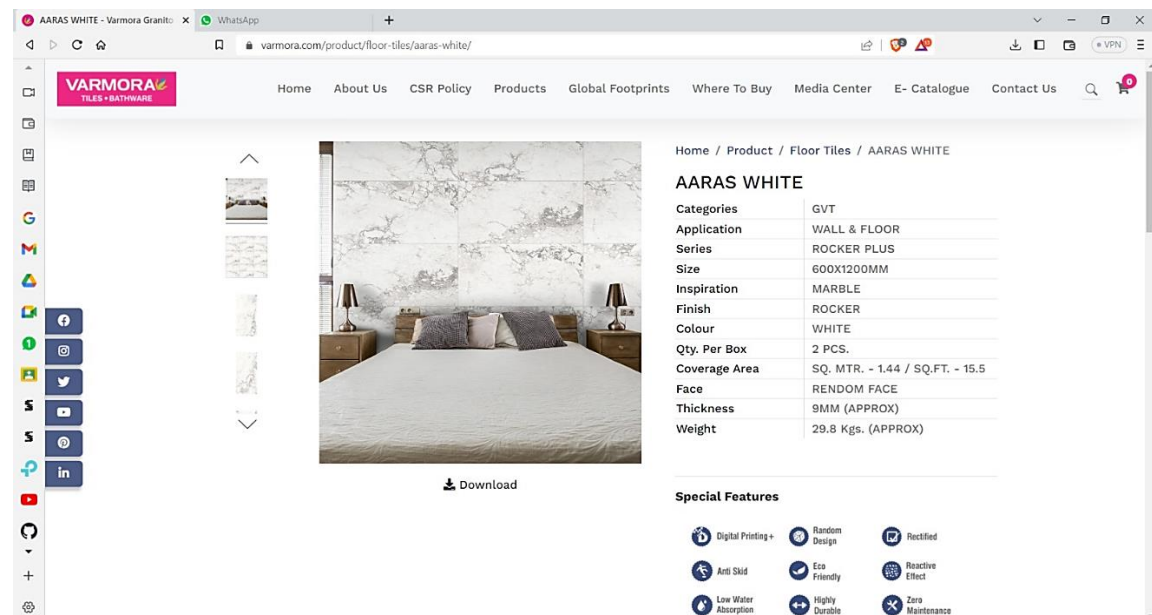
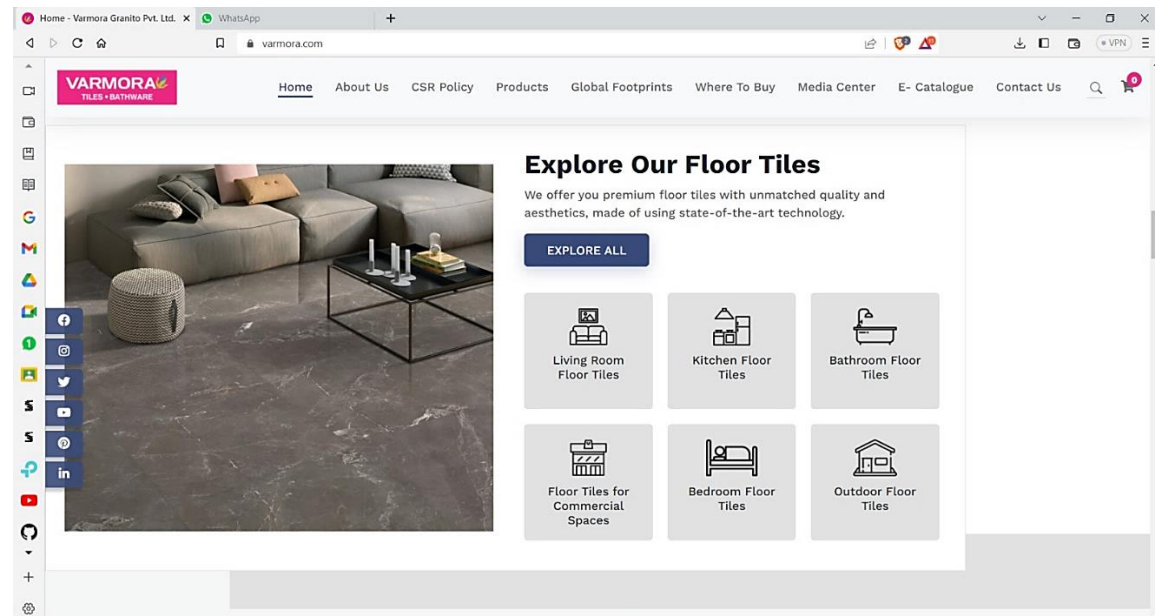
The Agile methodology is a way to manage a project by breaking it up into several phases. It involves constant collaboration with stakeholders and continuous improvement at every stage. Once the work begins, teams' cycle through a process of planning, executing, and evaluating.

The new system was build using this methodology so as inculcate the requirements of the new system in a systematic manner and not invasive in the already built code. This method provided a lot of ease as changes in the definitions were easily incorporated into the design logic implementation.

4.2 SAMPLES OF FORMS, REPORTS, AND INTERFACE

The following sections show the interface design of the various platforms used.

Collected data from the below website :



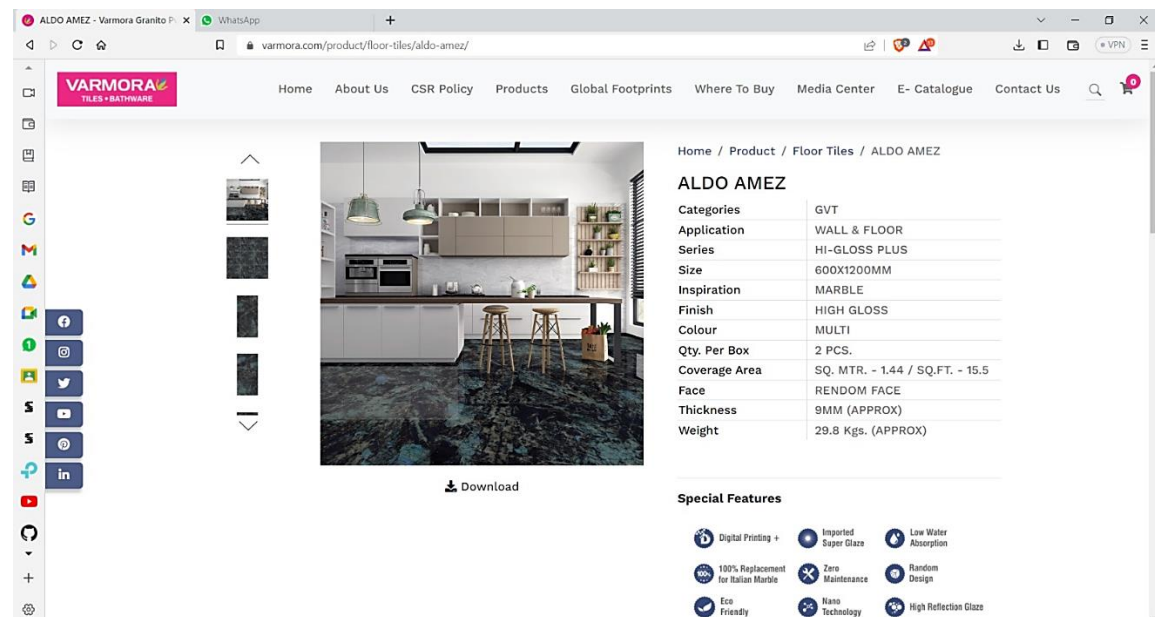


Fig 4.2.1 Website of Collected Data

Collected Data :

Fig 4.2.2 Collected Data

Database :

The figure consists of two screenshots of the Firebase console interface. The top screenshot shows the 'Realtime Database' section. The left sidebar contains the 'Project Overview' and 'Product categories' menu. The main content area displays the 'Realtime Database' with tabs for 'Data', 'Rules', 'Backups', 'Usage', and 'Extensions'. The 'Data' tab is active, showing a JSON structure for a 'recommendation' node. The structure includes a 'buttonValue' node with 'Input: 0' and 'flag: "false"', and a 'values' node with 'initial: 234' and 'recommendation: 302'. The URL bar shows 'https://project-e3a05-default-rtdb.asia-southeast1.firebaseio.com'. The bottom screenshot shows the 'Storage' section. The left sidebar is the same. The main content area displays the 'Storage' section with tabs for 'Files', 'Rules', 'Usage', and 'Extensions'. The 'Files' tab is active, showing a list of files. The URL bar shows 'https://project-e3a05.appspot.com/files'. The file list contains two entries: 'data_600.csv' (31.2 KB, text/csv, 13 Apr 2023) and 'tiles_data_full.csv' (106.58 KB, text/csv, 28 Mar 2023). Both screenshots include an 'Activate Windows' watermark in the bottom right corner.

Realtime Database

https://project-e3a05-default-rtdb.asia-southeast1.firebaseio.com

```
https://project-e3a05-default-rtdb.asia-southeast1.firebaseio.com/  
└─ recommendation  
  └─ buttonValue  
    └─ Input: 0  
    └─ flag: "false"  
  └─ values  
    └─ initial: 234  
    └─ recommendation: 302
```

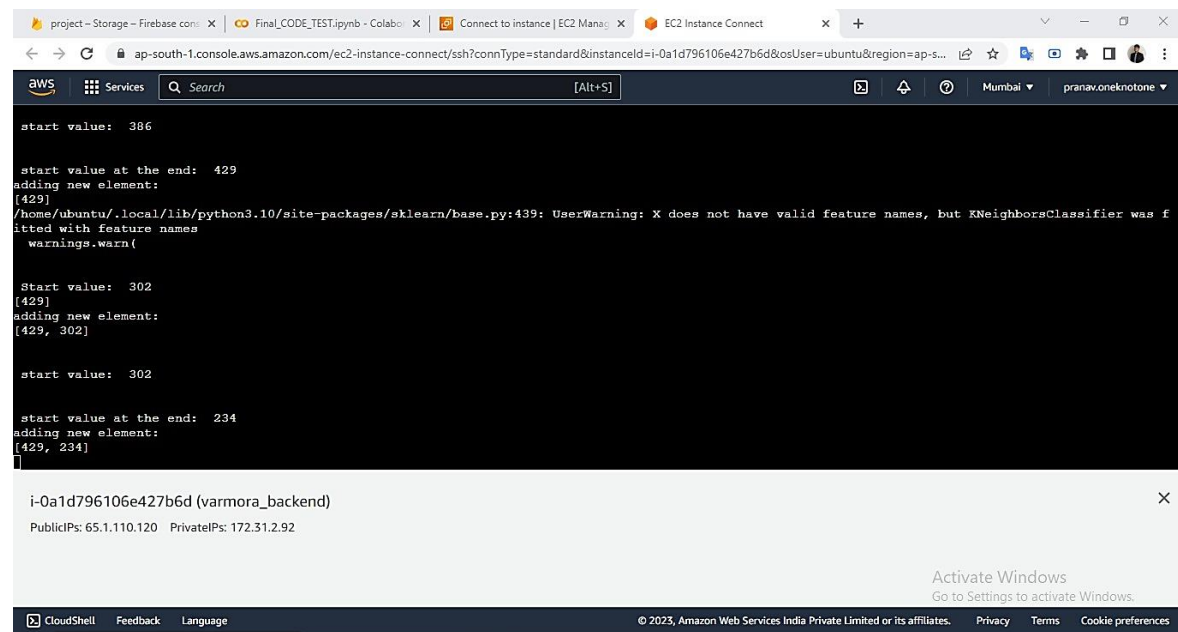
Database location: Singapore (asia-southeast1)

Storage

gs://project-e3a05.appspot.com

Name	Size	Type	Last modified
data_600.csv	31.2 KB	text/csv	13 Apr 2023
tiles_data_full.csv	106.58 KB	text/csv	28 Mar 2023

Fig 4.2.3 Database

AWS :


```

start value: 386

start value at the end: 429
adding new element:
[429]
/home/ubuntu/.local/lib/python3.10/site-packages/sklearn/base.py:439: UserWarning: X does not have valid feature names, but KNeighborsClassifier was fitted with feature names
  warnings.warn(

Start value: 302
[429]
adding new element:
[429, 302]

start value: 302

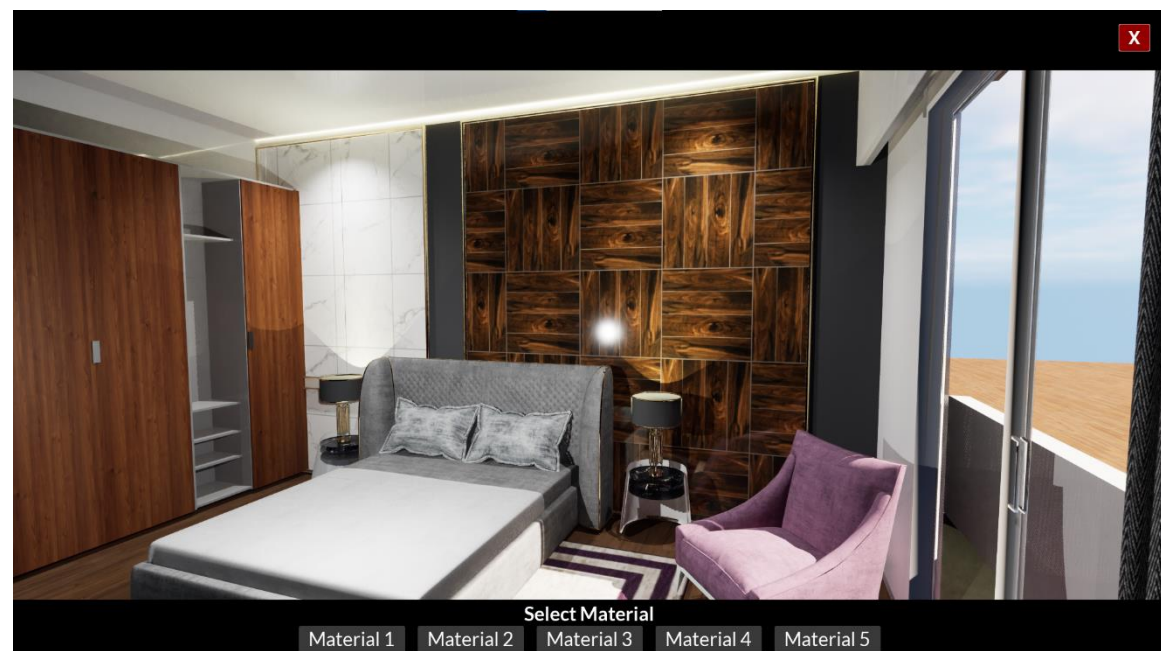
start value at the end: 234
adding new element:
[429, 234]

```

i-Oa1d796106e427b6d (varmora_backend)
PublicIPs: 65.1.110.120 PrivateIPs: 172.31.2.92

Activate Windows
Go to Settings to activate Windows.

CloudShell Feedback Language © 2023, Amazon Web Services India Private Limited or its affiliates. Privacy Terms Cookie preferences

Fig 4.2.4 AWS**Virtual Reality (User) :**

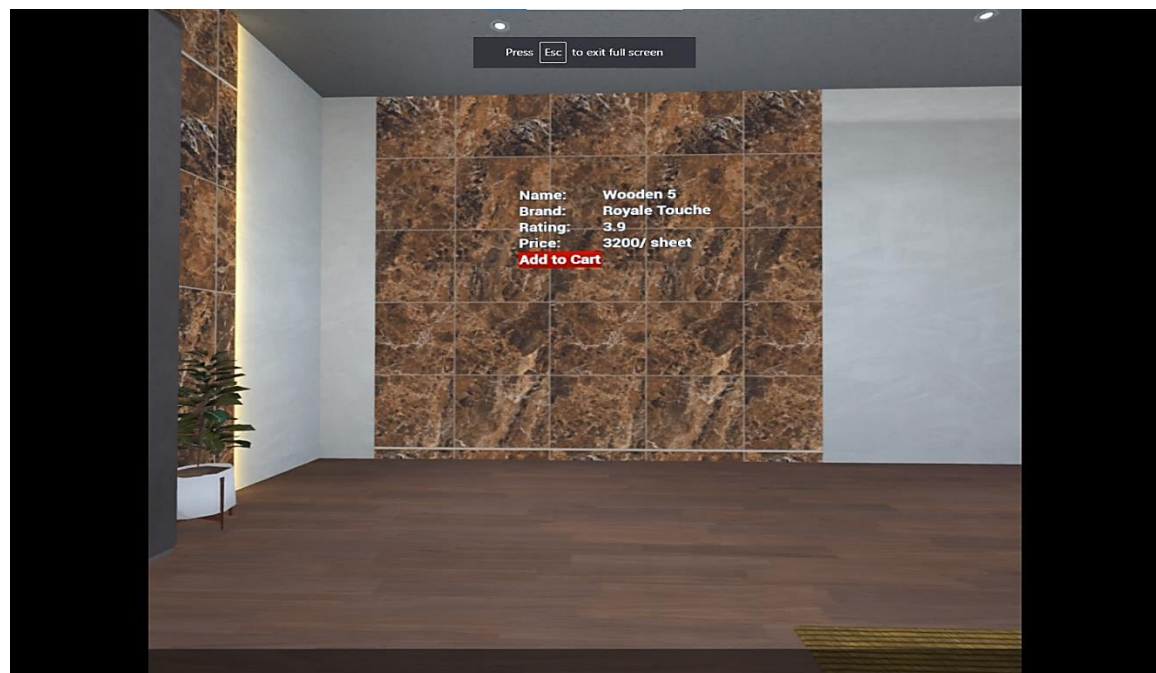
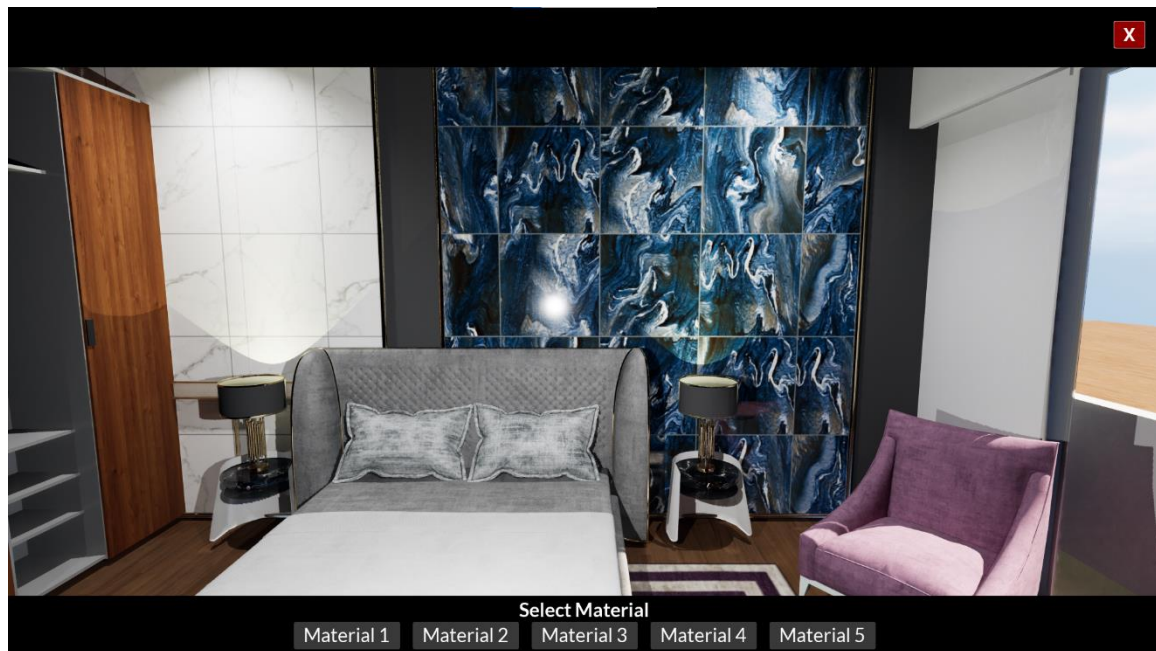




Fig 4.2.5 Virtual Reality (User)

CHAPTER 5 TESTING

5.1 TESTING PLAN / STRATEGY

Testing is an important aspect of building a web application. Here the system is tested based on some predefined test to see if the required output is there or not. There are numerous testing strategies, of which the ones adopted while building the application are:

1. Black Box Testing

The technique of testing without having any knowledge of the interior workings of the application is called black-box testing. The tester is oblivious to the system architecture and does not have access to the source code. Typically, while performing a black-box test, a tester will interact with the system's user interface by providing inputs and examining outputs without knowing how and where the inputs are worked upon.

2. White Box Testing

White-box testing is the detailed investigation of internal logic and structure of the code. White-box testing is also called glass testing or open-box testing. In order to perform white-box testing on an application, a tester needs to know the internal workings of the code. The tester needs to have a look inside the source code and find out which unit/chunk of the code is behaving inappropriately.

3. System Testing

System testing tests the system. Once all the components are integrated, the application is tested rigorously to see that it meets the specified Quality Standards. This type of testing is performed by a specialized testing team.

4. Functional Testing

We performed functional testing to verify that a software application performs and functions correctly according to design specifications. During functionality testing we check the core application functions, text input, menu functions and installation and setup on localized machines, etc.

CHAPTER 6 CONCLUSION AND DISCUSSION

6.1 OVERALL ANALYSIS OF INTERNSHIP LIABILITIES

The main objectives of the project in the internship were to learn about the different aspects that go into developing professional software good quality.

Given the problem statements that were to be executed in the project, there were many challenges that were faced but they were overcome with help from mentor. Everything in the process went on time, maybe a few setbacks, but then things got on track easily.

To count this internship as a success, there are several things that can be considered.

Some of which are:

1. Timely delivery of the complete project
2. Good quality of code
3. Working code with no clutter
4. Separate front-end and back-end code to maintain structure.

6.2 PROBLEM ENCOUNTERED AND POSSIBLE SOLUTIONS

Problems that were encountered during the internship period were:

1. Errors while collecting data due to uneven data entered on website. It was resolved by carefully analyzing the data and making required changes manually.
2. Selecting the best model to create the best recommendation model. It was resolved after understanding that it is an incremental process and is dependent on model's understanding of data so we planned on trying multiple models such as KNN, Kdtree, Neural Network etc.
3. Connecting database to the script due to our lack of experience. It was solved by carefully learning it and implementing it.
4. Hosting the script on AWS due to few technical error. It was resolved by help from a professional who had appropriate experience with it.

6.3 SUMMARY OF INTERNSHIP WORK

Being the team leader it was my responsibility to deliver product of the best quality and also collaborating with other teams to understand their way of thinking. This has helped me improve as a person and increase my management skills. However, this has also helped me in understanding the areas that need improvement.

This 3-month internship has been a successful journey in exploring new horizons and things that were not expected before. The confidence that is gained is enormous in terms of how well the code can be written.

In this duration soft skills were also polished as there were regular meetings with my mentor which helped in realizing how important communication is to maintain a simple logic channel between the parties to ensure that there are no discrepancies in understanding what the other has to say.

6.4 LIMITATION AND FUTURE ENHANCEMENT

Given that the project successfully implements the given problem statements from client, there are a few things which can be worked upon:

1. There are still ways of improving the recommendation system with better models and a more elaborate dataset.
2. The connection between the front-end and the back-end can be improved to reduce the points of failure.
3. The performance of the feature can be improved to allow multiple users to use it simultaneously while recommending products on their individual preferences.

REFERENCES

- [1] <https://www.selenium.dev/documentation/>
- [2] <https://beautiful-soup-4.readthedocs.io/en/latest/>
- [3] <https://scikit-learn.org/stable/>
- [4] <https://towardsdatascience.com/recommendation-systems-explained-a42fc60591ed>
- [5] <https://towardsdatascience.com/knn-algorithm-what-when-why-how-41405c16c36f>
- [6] <https://towardsdatascience.com/modern-recommendation-systems-with-neural-networks-3cc06a6ded2c>
- [7] <https://firebase.google.com/docs>